

Spring Rest HandsOn

1. pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.3</version>
  </parent>

  <groupId>com.cognizant</groupId>
  <artifactId>spring-learn</artifactId>
  <version>0.0.1-SNAPSHOT</version>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-devtools</artifactId>
    </dependency>

  </dependencies>

</project>

. application.properties

logging.level.org.springframework=info
logging.level.com.cognizant.springlearn=debug

logging.pattern.console=%d{yyMMdd} | %d{HH:mm:ss.SSS} | %-20.20thread | %5p | %-
```

```
25.25logger{25}|%25M|%m%n
```

```
server.port=8080
```

3. date-format.xml

Create inside **src/main/resources**

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
    <bean id="dateFormat"
        class="java.text.SimpleDateFormat">

        <constructor-arg value="dd/MM/yyyy"/>

    </bean>
```

```
</beans>
```

4. country.xml

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="
http://www.springframework.org/schema/beans
https://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<!-- Exercise 4 -->
```

```
<bean id="country"
    class="com.cognizant.springlearn.Country">

    <property name="code" value="IN"/>
    <property name="name" value="India"/>
```

```
</bean>
```

```
<!-- Exercise 6 -->
```

```
<bean id="in"  
    class="com.cognizant.springlearn.Country">  
    <property name="code" value="IN"/>  
    <property name="name" value="India"/>  
</bean>
```

```
<bean id="us"  
    class="com.cognizant.springlearn.Country">  
    <property name="code" value="US"/>  
    <property name="name" value="United States"/>  
</bean>
```

```
<bean id="de"  
    class="com.cognizant.springlearn.Country">  
    <property name="code" value="DE"/>  
    <property name="name" value="Germany"/>  
</bean>
```

```
<bean id="jp"  
    class="com.cognizant.springlearn.Country">  
    <property name="code" value="JP"/>  
    <property name="name" value="Japan"/>  
</bean>
```

```
<bean id="countryList"  
    class="java.util.ArrayList">
```

```
    <constructor-arg>  
        <list>  
            <ref bean="in"/>  
            <ref bean="us"/>  
            <ref bean="de"/>  
            <ref bean="jp"/>  
        </list>  
    </constructor-arg>
```

```
</bean>
```

</beans>

5. Country.java

```
package com.cognizant.springlearn;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

public class Country {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(Country.class);

    private String code;
    private String name;

    public Country() {
        LOGGER.debug("Inside Country Constructor.");
    }

    public String getCode() {
        LOGGER.debug("Inside getCode()");
        return code;
    }

    public void setCode(String code) {
        LOGGER.debug("Inside setCode()");
        this.code = code;
    }

    public String getName() {
        LOGGER.debug("Inside getName()");
        return name;
    }

    public void setName(String name) {
        LOGGER.debug("Inside setName()");
        this.name = name;
    }
}
```

```

@Override
public String toString() {
    return "Country [code="
        + code +
        ", name="
        + name +
        "]";
}
}

```

6. SpringLearnApplication.java

```

package com.cognizant.springlearn;

import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.List;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

@SpringBootApplication
public class SpringLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(
            SpringLearnApplication.class);

    public static void main(String[] args)
        throws Exception {

        SpringApplication.run(
            SpringLearnApplication.class,
            args);

        displayDate();
    }
}

```

```

        displayCountry();

        displayCountries();
    }

    public static void displayDate()
        throws Exception {

        LOGGER.info("START");

        ApplicationContext context =
            new ClassPathXmlApplicationContext(
                "date-format.xml");

        SimpleDateFormat format =
            context.getBean(
                "dateFormat",
                SimpleDateFormat.class);

        Date date =
            format.parse("31/12/2018");

        LOGGER.debug("Date : {}", date);

        LOGGER.info("END");
    }

    public static void displayCountry() {

        LOGGER.info("START");

        ApplicationContext context =
            new ClassPathXmlApplicationContext(
                "country.xml");

        Country country =
            context.getBean(
                "country",
                Country.class);

        Country anotherCountry =

```

```
        context.getBean(
            "country",
            Country.class);

    LOGGER.debug(
        "Country : {}",
        country);

    LOGGER.debug(
        "Another Country : {}",
        anotherCountry);

    LOGGER.info("END");
}

@SuppressWarnings("unchecked")
public static void displayCountries() {

    LOGGER.info("START");

    ApplicationContext context =
        new ClassPathXmlApplicationContext(
            "country.xml");

    List<Country> countries =
        (List<Country>)
        context.getBean(
            "countryList");

    LOGGER.debug(
        "Countries : {}",
        countries);

    LOGGER.info("END");
}
```

- spring-learn [boot] [devtools]
 - src/main/java
 - com.cognizant.springlearn
 - Country.java
 - SpringLearnApplication.java
 - src/main/resources
 - src/test/java
 - JRE System Library [JavaSE-21]
 - Maven Dependencies
 - target/generated-sources/annotations
 - target/generated-test-sources/test-annotations
 - src
 - target
 - pom.xml



